

Arcana Siege: Balance & Economy

Design workbook by Umut Eren Demirci. Blue cells are inputs; everything else is calculated or measured.

Sheet	What it contains	Live formulas?
Curve	Win rate of three bots (Random, Casual, Greedy) at six paces, against the design targets, with a chart.	Target checks
Where Runs End	Which wave broke the gate, for each bot's losses at 2.2 s a move.	Shares
Cards	The worth of each of the 16 upgrade cards: win rate when held from wave 1, and the change it makes.	No
Enemies & Towers	Enemy stats and tower damage; hits to kill, walk time and leaks to lose, per enemy.	Yes
Economy & Meta	Gold per siege by bot, and the price curve for permanent tower levels (ten per tower).	Yes
Campaign	The 12 sieges in 4 realms: health scale, boss, win bonus, and the career bot's tries to win each one.	Power ratio
Tuning	Every number in SiegeTuning, as the build uses it.	No

Source of truth: the game's code (Assets/_Project/Scripts/Siege/Core). Measured data comes from the balance runner (Tools/sim), which compiles those same rules and plays 250 sieges per cell, 500 per card, and 100 whole careers for the campaign. The story of the tuning is in Docs/BALANCE.md.

Win rate by bot and pace

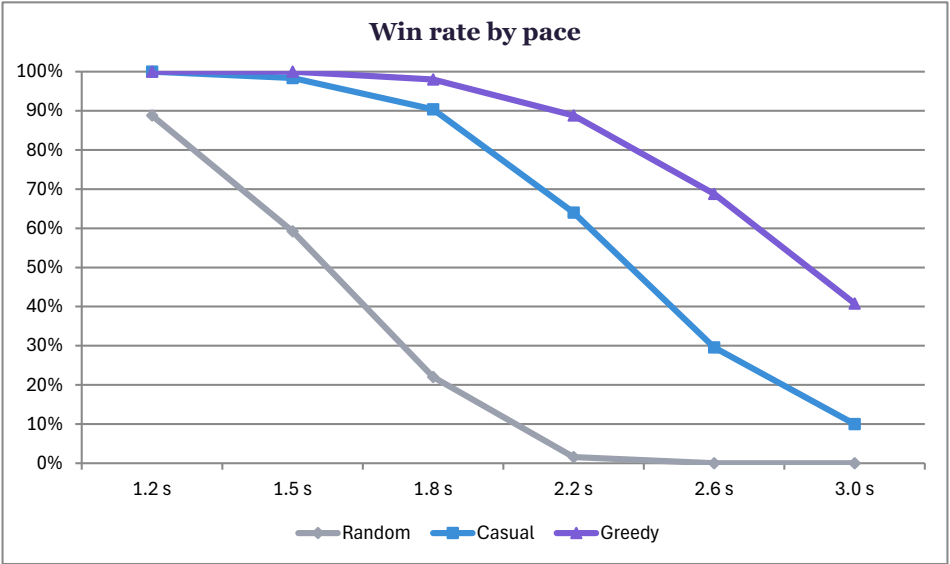
Pace = seconds per move, thinking plus animation. 250 sieges per cell.

Bot	1.2 s	1.5 s	1.8 s	2.2 s	2.6 s	3.0 s
Random	89%	59%	22%	2%	0%	0%
Casual	100%	98%	90%	64%	30%	10%
Greedy	100%	100%	98%	89%	69%	41%

Targets

Player	Bot	Pace	Low	High	Measured	On target?
First-time player	Casual	2.2 s	55%	65%	64%	Yes
Practised player	Greedy	1.8 s	90%	100%	98%	Yes
Practised but slow	Greedy	2.6 s	55%	70%	69%	Yes
Ignores the road	Random	1.8 s	0%	25%	22%	Yes

Reading the road is worth almost half your speed: the Greedy bot at 2.2 s a move plays as well as random swiping at 1.2 s. Mashing still wins, which is why armoured enemies were added: fire and tide spam is wasted on them.

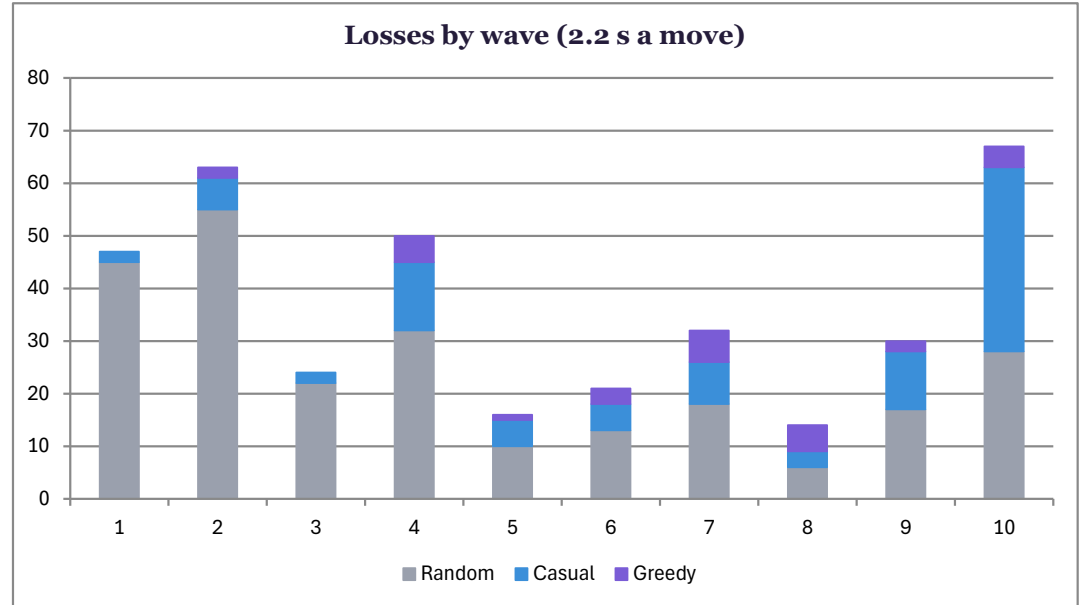


Where runs end

For the losses at 2.2 s a move: which wave broke the gate (250 sieges per bot).

Wave	Random	Casual	Greedy	Casual share
1	45	2	0	2%
2	55	6	2	7%
3	22	2	0	2%
4	32	13	5	14%
5	10	5	1	6%
6	13	5	3	6%
7	18	8	6	9%
8	6	3	5	3%
9	17	11	2	12%
10	28	35	4	39%
Total	246	90	28	

Wave 10 is the Devil: a boss that takes the largest share of losses feels earned. Waves 2 and 4 (runner waves) were early walls for first-time players and were thinned; wave 4 is still the biggest early loss.



Card power

The Greedy bot at 2.6 s a move starts the siege holding one card (500 sieges each). Differences under ~3 points are noise.

Card	Suit	Copies	Win rate	Change (points)
Executioner	Swords	2	78%	+12
Deep Water	Cups	1	77%	+11
Undertow	Cups	2	76%	+10
Ace of Cups	Cups	2	76%	+10
Rising Moon	Moon	2	75%	+9
Stone Skin	Pentacles	2	74%	+8
Bulwark	Pentacles	3	73%	+7
Ace of Swords	Swords	3	73%	+7
Twin Edges	Swords	1	71%	+6
Eclipse	Moon	2	71%	+5
Spreading Flame	Wands	2	70%	+4
Wildfire	Wands	2	69%	+3
Ace of Wands	Wands	3	69%	+3
Ace of Pentacles	Pentacles	3	69%	+3
The Wheel Turns	Sun	2	68%	+2
The Sun's Tithe	Sun	2	66%	-0

Normal draft (baseline)

66%

No draft at all

4%

-61.8

No card is a must-pick and none is dead. The Sun's Tithe pays gold, which only buys re-deals inside a run; its value arrives with the permanent upgrades of phase 5. The draft carries a run: without it the same bot wins a few percent.

Enemies and towers

Change a blue cell and every derived number updates. Values from SiegeTuning.cs.

Towers (damage per tile cleared)

Tower	Per tile	Match of 3	Match of 5
Fire (splash + burn)	6	18	30
Tide (whole road, slows)	2	6	10
Blade (front enemy)	14	42	70
Ward (repair)	7.5	22.5	37.5

Road length

20

Gate health

100

Enemies

Enemy	Health	Speed	Leak damage	Armour	Blade matches of 3 to kill	Fire matches of 3 to kill	Seconds to walk the road	Leaks that break the gate
Grunt	108	1.1	30	0%	3	6	18.2	4
Runner	58	2	18	0%	2	4	10.0	6
Brute	432	0.65	72	60%	11	60	30.8	2
Shieldbearer (shield)	90	1	30	0%	3	5	20.0	4
Acolyte (heals)	84	0.9	24	0%	2	5	22.2	5
Broodmother (splits)	300	0.7	48	0%	8	17	28.6	3
Imp	26	2.3	10	0%	1	2	8.7	10
The Devil (boss)	2917	0.35	1000	60%	70	406	57.1	1

Base health; each siege multiplies it by its scale (Campaign sheet), so the Devil has 3,500 in the Devil's Thicket (x1.2). Armour reads straight off the table: a Brute takes 11 Blade matches of three but 60 of Fire, so fire spam is wasted on it. A Shieldbearer's first direct hit is absorbed whatever its size; an Acolyte mends 12% of max health to everything within 3 units every 2.5 s; a Broodmother bursts into three Imps where it falls. The Arcana walk the road in about a minute and break the gate on arrival.

Economy and the meta

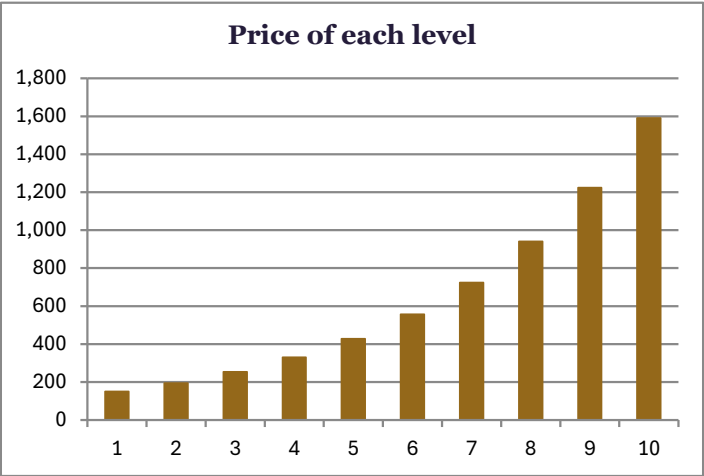
Gold per siege is measured at 1.8 s a move; prices as shipped in phase 5, with live formulas.

Bot at 1.8 s	Gold per siege	Spent on re-deals	Ultimates	Special moves	Moves
Random	319	0	13.5	23.9	134.0
Casual	536	0	19.5	57.1	151.9
Greedy	547	23	15.4	68.6	140.3

Permanent tower levels

First level costs	150
Each level costs x	1.3
Levels per tower	10
Towers	4
Gold per siege (Casual)	536

Level	Cost	Total for one tower	Total for all towers	Sieges to afford all
1	150	150	600	1.1
2	195	345	1,380	2.6
3	254	599	2,396	4.5
4	330	929	3,716	6.9
5	428	1,357	5,428	10.1
6	557	1,914	7,656	14.3
7	724	2,638	10,552	19.7
8	941	3,579	14,316	26.7
9	1,224	4,803	19,212	35.9
10	1,591	6,394	25,576	47.8



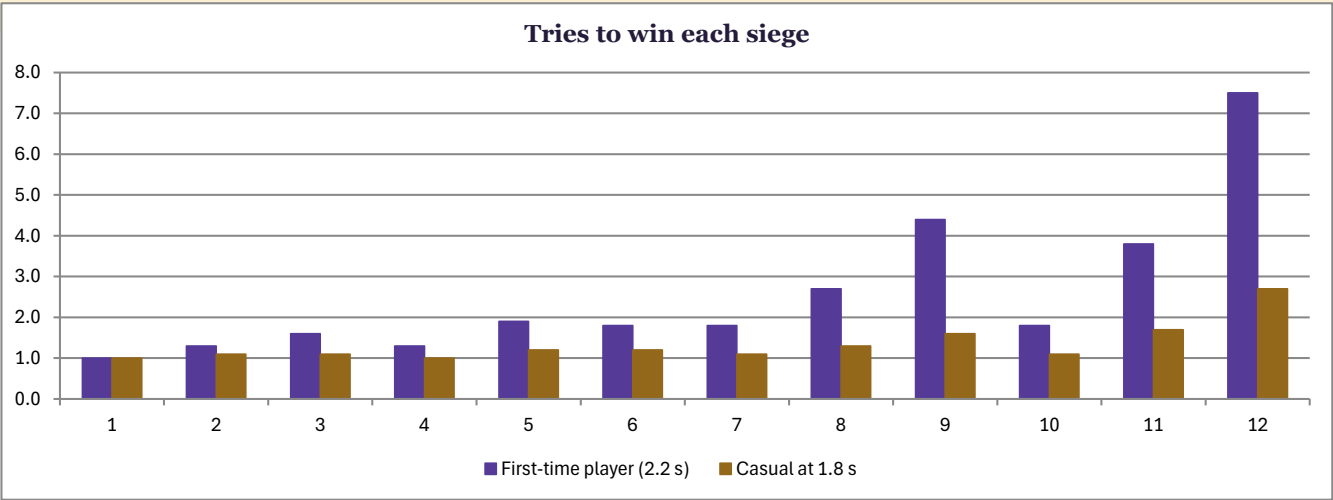
Inside a run gold only buys re-deals, so a siege ends with ~500 unspent gold; between runs it buys permanent tower levels (+12% output each). With three sieges, five levels at x1.5 were enough; the twelve-siege campaign needed a longer, gentler curve: ten levels at x1.3, about 6,400 gold per tower. Win bonuses (80 to 500 gold, Campaign sheet) keep a winning player a little ahead of the curve.

The campaign

Twelve sieges in four realms. Tries: runs until the siege first falls, from 100 careers of the career bot (Tools/sim/run.ps1 career).

#	Siege	Realm	Ends with	Enemy health x	Boss health	Win bonus	Tries: first-time 2.2 s	Tries: casual 1.8 s	Tries: practised 1.8 s	Levels owned at the win (of 40)	Tower power x	Health / power
1	The Fool's Gate	Greenwood	Old Gnasher (champion)	0.85	820	80	1.0	1.0	1.0	2.6	1.08	0.79
2	The Magician's Ford	Greenwood	The Ford Warlord (champion)	1.05	1,600	100	1.3	1.1	1.0	6.1	1.18	0.89
3	The Devil's Thicket	Greenwood	The Devil	1.20	3,500	150	1.6	1.1	1.0	9.8	1.29	0.93
4	The Chariot's Dunes	Sunscorched Road	The Shield Captain (champion)	1.35	2,400	150	1.3	1.0	1.0	12.4	1.37	0.98
5	The Hermit's Well	Sunscorched Road	The Sand Colossus (champion)	1.50	2,900	170	1.9	1.2	1.0	15.4	1.46	1.03
6	The Tower's Road	Sunscorched Road	The Tower	1.60	5,600	220	1.8	1.2	1.1	18.1	1.54	1.04
7	The Hanged Mire	Moonlit Marsh	The High Acolyte (champion)	1.75	3,300	220	1.8	1.1	1.1	20.2	1.61	1.09
8	The Star's Fen	Moonlit Marsh	The Bog Brute (champion)	1.80	3,500	240	2.7	1.3	1.1	22.5	1.68	1.07
9	The Moon's Keep	Moonlit Marsh	Death	1.85	6,300	300	4.4	1.6	1.1	25.8	1.77	1.04
10	The Empress's Pass	Frozen Summit	The Brood Queen (champion)	2.05	4,200	300	1.8	1.1	1.1	27.3	1.82	1.13
11	The World's Edge	Frozen Summit	The Frost Warlord (champion)	2.10	4,500	320	3.8	1.7	1.2	29.5	1.89	1.11
12	Judgement Peak	Frozen Summit	Judgement	2.15	8,000	500	7.5	2.7	1.6	33.1	1.99	1.08

Tower power: the average tower's output multiplier from the levels a first-time player owns when the siege falls. Health / power stays between about 0.8 and 1.15: the permanent upgrades roughly keep pace with the enemies, and what is left is the player's own skill and the draft. The tries form a sawtooth on purpose: a champion siege after each realm's finale is a breather, each finale is a wall.



Tuning

Every field of SiegeTuning, as the build uses it.

Field	Value
RoadLength	20
GateHealth	100
GateDamageTaken	1
WaveBreak	4
CascadeBonus	0.25
Drafts	TRUE
Grunt	hp 108 / speed 1.1 / gate 30 / gold 1
Runner	hp 58 / speed 2 / gate 18 / gold 1
Brute	hp 432 / speed 0.65 / gate 72 / gold 3
Boss	hp 3120 / speed 0.35 / gate 1000 / gold 25
GoldPerKillBonus	0
FireDamagePerTile	6
FireRadius	2.5
BurnDpsPerTile	1.5
BurnDuration	3
TideDamagePerTile	2
TideSlowPerTile	0.08
TideSlowMax	0.6
TideSlowDuration	3
BladeDamagePerTile	14
BladeCleave	0
BladeVersusTough	1
WardHealPerTile	7.5
ManaPerTile	10
ManaMax	100
UltimateDamage	150
VolleyTiles	4
QuakeStun	1.5
QuakeDamage	20
StarfallDamage	45
VolleyMana	40
VolleyGold	8